

27/WED/MAY

Assignment-1

1) Ans:

S = "Python"

Print (S[0], S[-1], S[6])

Trace back (most recent call last):

Index Error: String index out of range.

2) Positive Indexing

Negative Indexing.

* Positive indexing starts from 0 at the beginning of the string and moves left to right (0, 1, 2, 3, ...)

* Negative indexing starts from (-1) at the end of the string and moves right to left (-1, -2, -3, ...)

S = "Hello"

Print (S[-2])

3) Ans:

S = "code"

Print (S[10])

Traceback (most recent call last):

Index Error: string index out of range

Print (S[1:10])

code

Direct indexing (S[10]): Python demands an exact element. If the index doesn't exist

It through index error.

Slicing (`s[1:10]`): python handles boundaries gracefully.

4.) Ans:

Indexing concept:

`s = "mississippi"`

`first_s = s.find('s')`

`last_s = s.find('s')`

`Print (first_s, last_s)`

5.) Ans:

`s = 'abcdefg'`

`Print (s[-1], s[-3], s[-6])`

`(s[-1], s[-3], s[-6])`

a b c d e f
-3 -2 -1

Python resolves negative indices by adding them to them to the length of the string
(`len(s) = 6`)

6.) Ans:

Syntax: `String [start : stop : step]`

Start: * The index where the slice begins
* Default to 0 if step is positive.

Stop: * The index where the slice ends.
* Default to the end of the string
if step is positive.

step:- The increment value determining how many characters to skip.

7) Ans:-

`s = 'Hello'`

`Print (s[0:5])`

Hello

`Print s[5:]` → index 5 goes all the way to the end.

`Print s[:5]` → Hello.

`Print s[:]` → Hello.

8) Ans:-

`s = "Python"`

`Print (s[4:])`

output : "" (An empty string)

A positive step moves from left to right.

9) Ans:-

`s = "PythonSlicing"`

`Print (s[-3:])`

output : "ing"

10) Ans:-

`s = "abcdefgh"`

`Print (s[::1])`

`Print (s[::2])`

`Print(S[1::2])`

`S[::1]` : Traverses the whole string normally returning every single character.

`S[::2]` : Starts at index 0 & every 2nd character (indices 0, 2, 4, 6)

`S[1:2]` : Starts at index 1 & return every 2nd character (indices 1, 3, 5, 7)

11) Ans:

Step Value : -1

`S = "Python"`

`reversed_S = S[::-1]`

`Print(reversed_S)`
"nohtyp"

12) Ans:

`S = "abcdefgh"`

`Print(S[::-1])`
hgfedcba

`Print(S[::-2])`

hgdb

`Print(S[6:1:-1])`

gfedc

13) Ans:

`S = "a1b2c3d4e5"`

`Print(S[1::2])`

"12345"

14. Ans:-

28/MAY/2026

S = "Python"

Print(S[5:0:-1])

nohty

Print(S[5::-1])

nohtyP

Print(S[0:-1])

nohty

Starting number start with small value and stop with big value."

the positive number and negative number (are) using in slicing.

right to left are left to right it can be use.

15. Ans:-

S = "hello"

Print(S[100::-1])

"olleh"

Print(S[:: -100])

"o"

when start or stop values are bigger then the string length with negative step

python on will not gives error.

It will automatically adjust the

16) Ans:

S = "a b c d e f g h i j" len(S) = 12

middle_portion = S[2:-2]

print (middle_portion)

c d e f g h

Extract characters from index 2 to
second - last

17) Ans:

S = "racecar"

is_palindrome = S[::-1]

print (is_palindrome)

True.

18) Ans:

S = "Hello" len(S) = 5

print(S[1:100])

"ello"

print(S[-100:3])

"hell"

print(S[100:200])

Syntax Error.

why no error: exceed string boundaries

19) Ans:

```
S = "Hello world"
```

```
S = S[:6] + "python."
```

```
print(s)
```

```
"Hello python"
```

```
# Replace "world" with "python".
```

20) Ans:

```
S = "abcdefghij"
```

```
my_slice = slice(1, 8, 2)
```

```
print(s[my_slice])
```

```
bd fh.
```

```
# a b c d e f g h i j  
  0 1 2 3 4 5 6 7 8 9
```

```
# (start, stop, step)
```

```
This function standalone slice object
```

_____ x _____ x _____

assignment 1.py - C:\Python314\assignment 1.py (3.14.3)

File Edit Format Run Options Window Help

Python 3.14.3 (tags/v3.14.3:323c59a, Feb 3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)] on win32

Enter "help" below or click "Help" above for more information.

```
s="python"
```

```
print(s[0],s[-1],s[6])
```

Traceback (most recent call last):

File "<pyshell#1>", line 1, in <module>

```
print(s[0],s[-1],s[6])
```

IndexError: string index out of range

```
s="Hello"
```

```
print(s[-2])
```

```
1
```

```
s="code"
```

```
print(s[10])
```

Traceback (most recent call last):

File "<pyshell#9>", line 1, in <module>

```
print(s[10])
```

IndexError: string index out of range

```
print(s[1:10])
```

```
ode
```

```
s="mississippi"
```

```
first_s=s.find('s')
```

```
last_s=s.find('s')
```



Q Search



ENG
IN



Ln: 1 Col: 0
11:08 AM
28-05-2026


```
s="mississippi"  
first_s=s.find('s')  
last_s=s.find('s')  
print(first_s,last_s)  
2 2
```

```
s='abcdef'  
print(s[-1],s[-3],s[-6])  
f d a
```

```
s='Hello'  
print([0:5])  
SyntaxError: invalid syntax  
print(s[5:1])  
  
print(s[5: ])  
  
print(s[ :5])  
Hello
```

```
print(s[ : ])  
Hello
```



Search



ENG
IN



```
s="python"  
print(s[4:1])
```

```
s="pythonslicing"  
print(s[-3: ])  
ing
```

```
s="abcdefgh"  
print(s[::1])  
abcdefgh  
print(s[::2])  
aceg  
print(s[1:2])  
b  
print(s[1::2])  
odfh
```

```
s="python"  
reversed_s=s[::-1]  
print(reversed_s)  
nohtyp
```

```
s="abcdefgh"
print(s[::-1])
hgfedcba
print(s[::-2])
hfdb
print(s[6:1:-1])
gfedc
```

```
s="a1b2c3d4e5"
print(s[1::2])
12345
```

```
s="python"
print(s[5:0:-1])
nohty
print(s[5::-1])
nohtyp
print(s[:0:-1])
nohty
```

```
s="python"
print(s[100::-1])
nohtyp
```

```
s="python"
print(s[100::-1])
nohtyp
```

```
s="hello"
print(s[100::-1])
olleh
print(s[::-100])
h
print(s[::100])
o
```

```
s="abcdefghij"
middle_portion=s[2:-2]
SyntaxError: invalid syntax
middle_portion=s[2:-2)
SyntaxError: closing parenthesis ')' does not match opening parenthesis '['
middle_portion=s[2:-2]
print(middle_portion)
cdefgh
```

```
s="racecar"
>>> is_palindrom=s[::-1]
>>> print(is_palindrom)
racecar
```


assignment 1.py - C:\Python314\assignment 1.py (3.14.3)

File Edit Format Run Options Window Help

```
s="racecar"
>>> is_palindrom=s[::-1]
>>> print(is_palindrom)
racecar
>>>
>>>
>>> s="hello"
>>> print(s[1:100])
ello
>>> print(s[-100:3])
hel
>>> Print(s[100:200])
Traceback (most recent call last):
  File "<pysHELL#95>", line 1, in <module>
    Print(s[100:200])
NameError: name 'Print' is not defined. Did you mean: 'print'?
>>>
>>>
>>> s="Hello World"
>>> s=s[:6] +"python"
>>> print(s)
Hello python
>>>
>>>
>>> s="abcdefghij"
>>> my_slice=slice (1,8,2)
>>> print(s[my_slice])
hdfh
```

Breaking news
Siddaramaiah to

Search



ENG
IN

Ln 22 C

11:10 AM
28-05-2024

assignment 1.py - C:\Python314\assignment 1.py (3.14.3)

File Edit Format Run Options Window Help

```
>>> is_palindrom=s[::-1]
```

```
>>> print(is_palindrom)
```

```
racecar
```

```
>>>
```

```
>>>
```

```
>>> s="hello"
```

```
>>> print(s[1:100])
```

```
ello
```

```
>>> print(s[-100:3])
```

```
hel
```

```
>>> Print(s[100:200])
```

```
Traceback (most recent call last):
```

```
File "<pyshell#95>", line 1, in <module>
```

```
Print(s[100:200])
```

```
NameError: name 'Print' is not defined. Did you mean: 'print'?
```

```
>>>
```

```
>>>
```

```
>>> s="Hello World"
```

```
>>> s=s[:6] +"python"
```

```
>>> print(s)
```

```
Hello python
```

```
>>>
```

```
>>>
```

```
>>> s="abcdefghij"
```

```
>>> my_slice=slice(1,8,2)
```

```
>>> print(s[my_slice])
```

```
odfh
```

Breaking news
US forces target

Search



Ln 22 Co

11:10 AM

28-05-2026